

Kenji Gunawan

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EDUCATION

University of California San Diego

Expected Grad: Spring 2027

B.S. Applied Mathematics and Data Science (Double Major)

GPA: 3.94/4.00 (Provost Honors)

Top 378 Putnam Qualifier (2023), Gold Medalist of the 2022 Mathematics Indonesian National Science Olympiad (OSN),

2nd place in USA for IMC Prosperity Challenge 2025 (9/12625 Global, Top 0.08%)

Activities: Data Science Student Society (DS3), Triton Quantitative Trading (TQT)

Selected Coursework: Principles of ML Algorithms, Probability (and measure-theoretic), Stochastic Processes I-II, Math. Stat. I, Optimization, OOP and DS in Java, Algorithms Analysis and Design, Honors Real Analysis I-II, Fourier Analysis, Practical EDA

EXPERIENCE

Quantitative Researcher Intern

Jakarta, Indonesia

Pinnacle Investment

Jun 2024 – Sep 2024

- Used **pandas**, **seaborn**, and **matplotlib** to develop strategies on U.S. equity swing-trading strategies, backtest systematically
- Built **web-scraping** tool (**Selenium**, **BeautifulSoup**, **pandas**) to extract alternative financial signals into structured datasets.
- Improved **ML data pipeline quality** by conducting structured statistical analysis (YoY correlation, quintile hypothesis tests) and XGBoost feature importance scoring, reducing input noise and **improving feature robustness** for the proprietary model.
- Prototyped **ARIMA** and tree-based regressors on macro indicators, assessing feasibility and limitations in recession forecasts.

PROJECTS

Team Alpha Animals: IMC Prosperity Challenge 2025 (Ended in 9th of 12625, 2nd in USA: Elite Contenders Award)

- Implemented a **Black-Scholes model** for pricing volcanic rock **options** using rolling IV, pushing **58th to 2nd** globally in R3.
- Performed proper PnL attribution: **Traced a 73,600 seashell profit** to a perpetual **short position**, preserving our edge in R4.
- Led manual trading rounds: **EV calculations**, discussed **position sizing** and **game-theory** to optimize risk-adjusted returns.
- Leveraged **backtester** and **visualizer** to dissect PnL, position, and order-book dynamics across **10+ products**, enabling **rapid strategy iterations** which led to reallocation of capital into **high-edge** voucher **trades** and arbitrage in Magnificent Macarons.

AppLovin Ad Intelligence (Feature Engineering) Challenge / PyTorch, NumPy, Scikit-learn (CalHacks 2025)

- Built a feature-engineering pipeline with ImageBind **transformer** and **performed PCA** to learn orthogonal latent features.
- Improved **explained variance** of the top principal axis from **~11 to ~26%** by integrating **OCR** signals in embedding pipeline.
- Selected factors through **cosine-similarity** activation profiles and **pairwise correlation** for interpretability and stability.

MarketPulse / YFinance, Pandas, HuggingFace FinBERT (**Most Innovative Award: LPL Financial Hackarama 2025**)

- Designed end-to-end pipeline** of news-driven prediction: spanning input processing, model inference, and user-facing output.
- Applied FinBERT for **named entity recognition** to extract company-linked keywords and rank financial news by relevance.
- Built a live **YFinance Scraper** to stream market data in sync with real-time news recommendations.

RESEARCH

Pivot Probabilities and Norm Effects in Gaussian Elimination for β -ensembles

La Jolla, CA

Co-authored with: Prof. John Peca-Medlin (Pre-print: <https://arxiv.org/abs/2506.20470>)

May 2025 – Jul 2025

- Derived the **exact pivot probability** for 2×2 GUE (Gaussian Unitary Ensemble) matrices under the **L^1 -norm** used by LAPACK, contrasting the L^2 -norm typically assumed by reducing to an integral over surface of a sphere via symmetry.

LEADERSHIP AND OUTREACH INITIATIVES

Marketing Team of **DataHacks 2025**

Nov 2024 – Apr 2025

- Created Instagram posts for @ds3atucsd's Instagram webpage, co-created the Squid Games-themed promotional videos.

Participant in the Season 1 of the viral Indonesian game show: **Ruangguru Clash of Champions**

May 2024 – Aug 2024

- In addition, it also grew my IG account to **over 85,000 followers**; producing short-form content tailored to Indonesian youth.

Mathematics Olympiad Coach and Problem Writer in **The Strikers Edu, Mandiri KSN Education,**

Pejuang OSN, and Sobat Arin

Dec 2021 – Dec 2023

- Taught **250+ students** advanced math (olympiad-level **combinatorics, number theory, algebra**); co-created original contests and lecture notes, with **one-third** of Indonesia's 2023 National Science Olympiad winners **coached by our team**.

General Coordinator of **Smakonecup 2022: JHS Competition**

Jan 2022 – Nov 2022

- Led **26 people** in the team, **increasing** participants by **over 100%** from previous year via coordinating a technical overhaul.

SKILLS

Tools/Skills: Python • Java • Jupyter • Excel • Git • Cursor • R • HTML • MATLAB • C • Canva • SQL • OpenAI API

Frameworks/Libraries: Pandas • Scipy • Seaborn • Selenium • Streamlit • Scikit-learn • Matplotlib • XGBoost • HuggingFace

Interests: Badminton, Pickleball, Hiking, Cooking, Culinary Exploration, Traveling, Singing, Journaling, Wrestling, Lacrosse